OL | Organic Letters

pubs.acs.org/OrgLett Letter

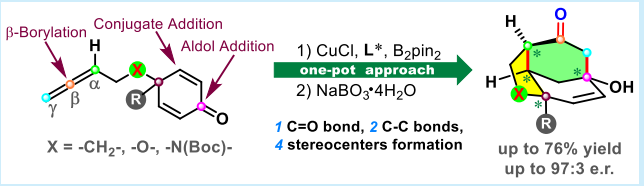
One-Pot Preparation of Tricyclo[5.2.2.0^{4,9}]undecanes via Cu-Catalyzed Asymmetric Carboboration of Cyclohexadienone-Tethered Allenes

Kai-Rui Feng,[#] Yun-Xuan Tan,[#] Wenbo Ye, Yi-Fan Wang, Zhi-Tao He,* Ping Tian,* and Guo-Qiang Lin*

Cite This: *Org. Lett.* 2021, 23, 607−611 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

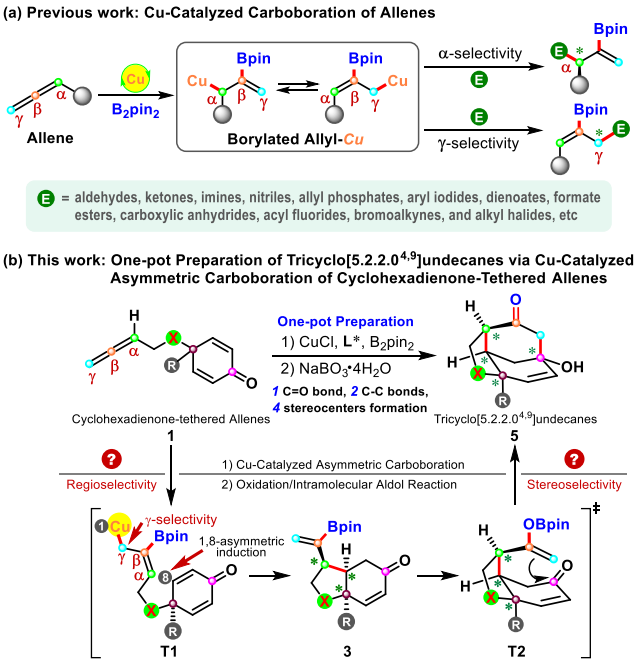
ABSTRACT: The Cu-catalyzed asymmetric carboboration of cyclohexadienone-tethered allenes has been achieved through regioselective β-borylation of the allenes and subsequent conjugate addition to cyclohexadienones, affording *cis*-bicyclic frameworks with acceptable yields and high to excellent enantioselectivities. Further conjugate borylation of the carboboration products proved to be a favorable kinetic resolution process, which improved the overall enantioselectivity. Finally, one-pot preparation of highly enantioenriched tricyclo[5.2.2.0^{4,9}]undecanes was developed from the cyclohexadienone-tethered allenes through β-borylation/1,4-addition and subsequent tandem oxidation/intramolecular aldol reaction.



Tricyclo[5.2.2.0^{4,9}]undecanes represent a unique framework existing in a number of natural products,[1] for instance, (+)-upial,[1a] JBIR-85,[1b] fusidilactone C,[1c] garcixanthochymone C,[1d] and shinjulactone C.[1e] Because of their structural complexity, efficient synthetic approaches to such intriguing bridged tricyclic skeletons are extremely rare. Recently, significant progress has been made in borylcopper (Cu−B)-catalyzed borylation of allenes,[2] especially carboboration of allenes (Scheme 1a).[3−9] The reactions of Cu−B species with allenes result in useful β-borylated allylcopper intermediates, which subsequently undergo a variety of transformations with different carbon electrophiles: aldehydes, ketones, imines, and nitriles for 1,2-addition;[3] allyl phosphates for boroallylation;[4] aryl iodides for boroarylation;[5] dienoates for 1,6-addition;[6] formate esters, carboxylic anhydrides, and acyl fluorides for boroacylation;[7] bromoalkynes for boroalkynylation;[8] and alkyl halides for alkylboration.[9] Despite the above notable advances, the 1,4-addition of β-borylated allylcopper especially to α,β-unsaturated ketones has never been uncovered,[10] due to the fact that the six-membered chairlike transition state makes the 1,2-addition to ketone more favorable.[3a] Herein, we present Cu-catalyzed asymmetric carboboration of cyclohexadienone-tethered allenes,[11] which involves asymmetric 1,4-addition of β-borylated allylcopper to enone, thereby establishing a one-pot preparation of highly enantioenriched tricyclo[5.2.2.0^{4,9}]undecanes via the aforementioned carboboration and subsequent tandem oxidation/intramolecular aldol reaction (Scheme 1b).

This one-pot process starts from the Cu-catalyzed regioselective β-borylation of the allene moiety as the direct conjugate borylation is restrained from the neighboring steric

Scheme 1. Strategic Design for the One-Pot Preparation of Tricyclo[5.2.2.0^{4,9}]undecanes



Received: December 18, 2020
Published: December 30, 2020

Downloaded via FUDAN UNIV on January 18, 2022 at 05:12:05 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

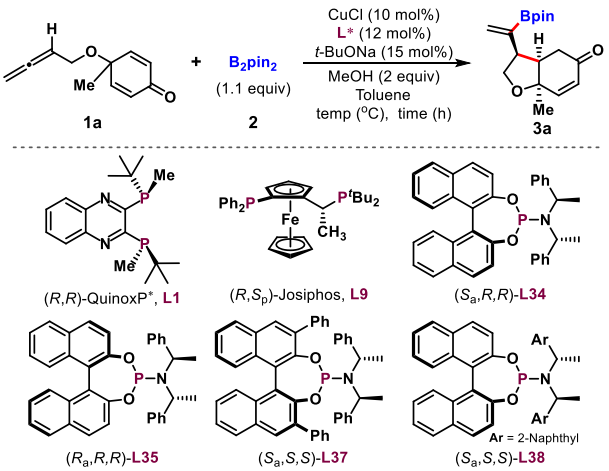
© 2020 American Chemical Society

https://dx.doi.org/10.1021/acs.orglett.0c04191
Org. Lett. 2021, 23, 607−611

congestion (Scheme 1b).[12] The Cu−B addition places the Cu at the less hindered site of allene. Both γ-selective copper addition and the highly reactive nature of cyclohexadienone could nicely prevent the competitive β-elimination pathway.[11a,13] Thus, the subsequent intramolecular 1,4-addition of **T1** affords chiral *cis*-bicyclic frameworks **3**. The next tandem oxidation/intramolecular aldol reaction furnishes the expected tricyclo[5.2.2.0^{4,9}]undecanes **5**, probably due to the close location of this enol boronate moiety to the ketone functionality in **T2**. However, the multiple control of overall regio-, diastereo-, and enantioselectivity (1,8-asymmetric induction) remains extremely challenging in these domino processes.

With these considerations in mind, a set of representative chiral ligands was evaluated for this Cu-catalyzed asymmetric carboboration of cyclohexadienone-tethered allene **1a**, and selected results were summarized in Table 1.[14] Initially, the conventionally used ligands (*R*,*R*)-QuinoxP* (**L1**) and (*R*,*S*$_p$)-Josiphos (**L9**) were tested.[15] Fortunately, the desired product **3a** was obtained, albeit in moderate diastereoselectivities and low enantioselectivities (Table 1, entries 1 and 2). Previously, we reported that phosphoramidite ligand **L34** could promote the asymmetric carboboration of 1,6-enynes.[16] However, only 20% yield and moderate er were observed despite the excellent diastereoselectivity in this case (Table 1, entry 3). Next, its diastereoisomeric ligand **L35** was subjected to the same reaction; to our delight, the er of **3a** dramatically rose to 88:12 (Table 1, entry 4). Attempt to increase the size of ligand **L35** to either **L37** or **L38** led to decreased er (Table 1, entries 5 and 6). Lowering the reaction temperature to −15 °C resulted in great improvement of er (Table 1, entries 7−9), and increasing B_2pin_2 (**2**) loading gave rise to further improvement of both yields and enantioselectivities (Table 1, entries 10−12). In particular, moderate yield (45%) and excellent diastereo- and enantioselectivity (>25:1 dr and 95:5 er) were accomplished when 2 equiv of B_2pin_2 was used (Table 1, entry 11). We also found that perfect enantioselectivity (>99.9:0.1 er), but with only 21% yield, was provided when more *t*-BuONa (30 mol %) was used (Table 1, entry 13). The low yield of **3a** might result from second conjugate borylation of excessive B_2pin_2 to **3a** promoted by extra *t*-BuONa.

Table 1. Selected Optimization Studies[a]



entry	L*	temp (°C)	time (h)	conv[b] (%)	yield (%)	dr	er[e]
1	**L1**	25	9	75	49[c]	17:1[d]	45:55
2	**L9**	25	9	85	53[c]	6:1[d]	64:36
3	**L34**	25	9	50	20[b]	20:1[b]	20:80
4	**L35**	25	14	80	40[b]	20:1[b]	88:12
5	**L37**	25	17	90	40[b]	20:1[b]	41:59
6	**L38**	25	17	75	40[b]	20:1[b]	15:85
7	**L35**	0	19	80	40[b]	20:1[b]	91:9
8	**L35**	−15	19	70	40[b]	20:1[b]	92:8
9	**L35**	−35	22	45	25[b]	20:1[b]	86:14
10[f]	**L35**	−15	21	85	47[c]	>25:1[d]	94:6
11[g]	**L35**	−15	11	90	45[c]	>25:1[d]	95:5
12[h]	**L35**	−15	21	90	53[c]	>25:1[d]	91:9
13[g,i]	**L35**	−15	13	95	21[c]	>25:1[d]	>99.9:0.1

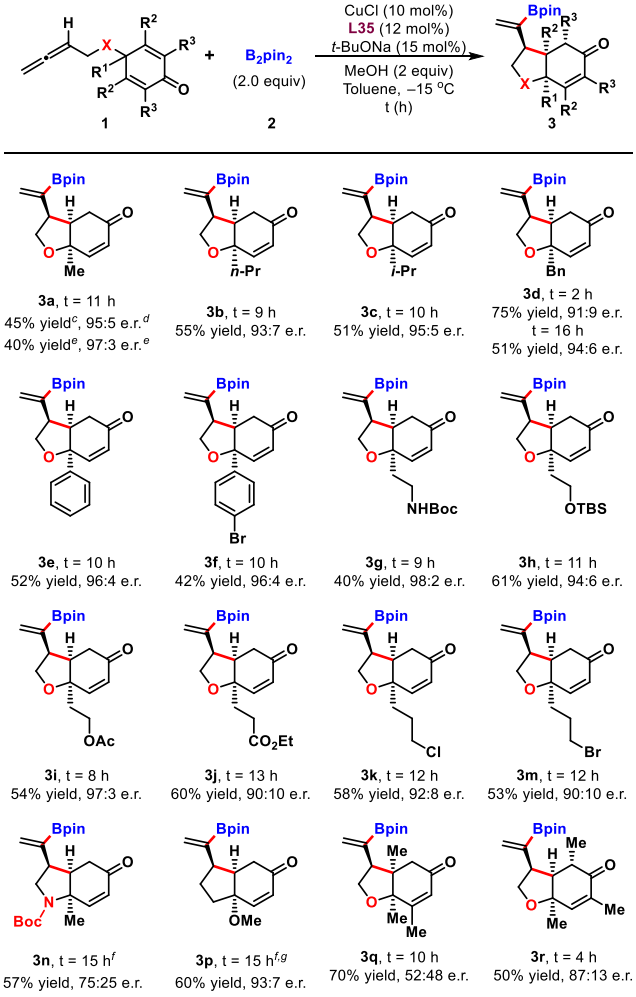
[a]Reactions were performed under an Ar atmosphere. [b]Determined by ^{1}H NMR of unpurified mixtures. [c]Yield of isolated product **3a**. [d]Determined by ^{1}H NMR of isolated product **3a**. [e]Determined by HPLC analysis. [f]B_2pin_2 (**2**, 1.5 equiv) was used. [g]B_2pin_2 (**2**, 2.0 equiv) was used. [h]B_2pin_2 (**2**, 3.0 equiv) was used. [i]*t*-BuONa (30 mol %) was used.

With the optimal conditions identified, we next evaluated the scope of cyclohexadienone-tethered allenes **1**. With the R^1 substituent as an alkyl, benzyl, or phenyl group, the reactions proceeded smoothly with moderate to good yields (42%−75%) and good to high enantioselectivities (91:9−96:4 er, Scheme 2, **3a**−**3f**). The absolute configuration of product **3e** was unambiguously determined as (3*R*,3a*R*,7a*R*) by X-ray diffraction analysis. Moreover, as for the benzyl substrate, prolonging the reaction time led to erosion of the yield but slight enhancement of the enantioselectivity (Scheme 2, **3d**), which might be attributed to the further conjugate borylation of excessive B_2pin_2 to **3d**. With a heteroatom (N, O, Cl, and Br) as part of the R^1 substituent in substrates **1**, the reaction yields and er values remained moderate and high, respectively (Scheme 2, **3g**−**3m**). It is worth mentioning that the highly reactive bromoalkyl group was comfortably tolerated without any observation of nucleophilic substitution product in the transformation (Scheme 2, **3m**).[9b] For the N- and C-linked substrates **1n** and **1p**, which were less reactive, the corresponding carboboration products, *cis*-hydroindole **3n** and *cis*-hydroindene **3p**, could also be obtained with moderate to high enantioselectivities at room temperature. For β-methyl-substituted cyclohexadienone substrate **1q**, almost racemic product **3q** was obtained in good yield, which was probably attributed to the steric effect in the conjugate addition process, whereas moderate yield and enantioselectivity could be achieved when α-methyl-substituted cyclohexadienone substrate **1r** was used (Scheme 2, **3r**). As shown in Scheme 2, the mild reaction conditions are generally compatible with a wide range of functional groups, such as amine, silyl ether, ester, chloride, and bromide, providing a versatile platform for further synthetic manipulations.

To confirm that the second conjugate borylation of carboboration products **3** was a favorable kinetic resolution process,[11b] i.e., the minor enantiomer of the carboboration product was consumed faster by excessive B_2pin_2 through second conjugate borylation, racemic product (±)-**3a** was subjected to the standard reaction conditions (Scheme 3). In line with our expectations, the conjugate borylation product **4a** was obtained with 47% yield and 86:14 er and the remaining **3a** was determined to be (−)-**3a** with 38% yield and 93:7 er (Scheme 3, eq 1). When more *t*-BuONa (30 mol %) was used, the yield of (−)-**3a** decreased to 30%; however, the loss was compensated by the achievement of excellent enantioselectivity (98:2 er, Scheme 3, eq 2). This study clearly demonstrated that

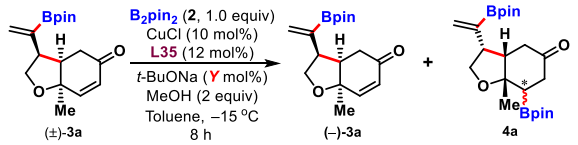
https://dx.doi.org/10.1021/acs.orglett.0c04191
Org. Lett. 2021, 23, 607−611

Scheme 2. Substrate Scope of Cyclohexadienone-Tethered Allenes[a]



[a]Reactions were performed under an Ar atmosphere. [b]All dr > 25:1 (determined by ^{1}H NMR analysis of unpurified mixtures). [c]Yield of isolated product **3**. [d]Determined by HPLC analysis. [e]1.0 mmol scale. [f]At room temperature. [g]**L35** was replaced by (*S*)-BINAP (**L3**).

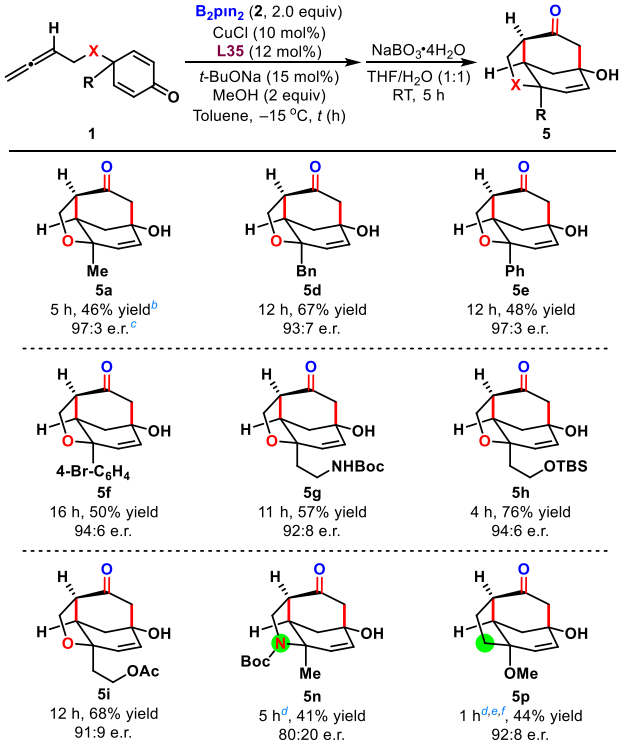
Scheme 3. Kinetic Resolution of (±)-3a via Cu-Catalyzed Conjugate Borylation



the second conjugate borylation of carboboration products was a favorable kinetic resolution process, and this can explain why perfect enantioselectivity of the carboboration reaction could be achieved by adding more base and excessive B_2pin_2 (Table 1, entry 13).

Finally, the one-pot preparation of highly enantioenriched tricyclo[5.2.2.0^{4,9}]undecanes was developed from the simple cyclohexadienone-tethered allenes, and the results are illustrated in Scheme 4.[17] With the R substituent as an alkyl,

Scheme 4. One-Pot Preparation of the Highly Enantioenriched Tricyclo[5.2.2.0^{4,9}]undecanes[a]



[a]Reactions were performed under an Ar atmosphere. [b]Yield of isolated product **5**. [c]Determined by HPLC analysis. [d]At room temperature. [e]B_2pin_2 (**2**, 1.2 equiv) was used. [f]**L35** was replaced by (*S*)-BINAP (**L3**).

benzyl, or phenyl group, the domino reaction proceeded smoothly to give tricyclo[5.2.2.0^{4,9}]-2-oxaundecanes with moderate yields (46%−67%) and high to excellent enantioselectivities (93:7−97:3 er, Scheme 4, **5a**, **5d**, **5e**, and **5f**). With a heteroatom (N and O) as part of R substituent in substrates **1**, the reaction yields and er values remained moderate and high, respectively (Scheme 4, **5g**, **5h**, and **5i**). For the N- and C-linked substrates, **1n** and **1p**, the corresponding tricyclo[5.2.2.0^{4,9}]undecanes were obtained with moderate to high enantioselectivities as well (Scheme 4, **5n** and **5p**). The relative configuration of products **5e** and **5f** was determined by the X-ray diffraction analysis of their racemates, respectively.

In summary, the first Cu-catalyzed asymmetric carboboration of cyclohexadienone-tethered allene has been successfully established. This tandem reaction proceeded smoothly to afford *cis*-hydrobenzofuran, *cis*-hydroindole, and *cis*-hydroindene frameworks bearing three consecutive carbon chiral centers with good yields and high to excellent enantioselectivities. Adding more base or excessive B_2pin_2 to the aforementioned reaction could increase the enantioselectivity. Such enhancement was mainly caused by the second conjugate borylation of extra B_2pin_2, which was supported by further kinetic resolution studies using racemic carboboration products. Furthermore, the one-pot preparation of highly enantioenriched tricyclo[5.2.2.0^{4,9}]undecanes was established directly from the simple cyclohexadienone-tethered allenes via domino processes: *β*-borylation/1,4-addition and subsequent tandem oxidation/intramolecular aldol reaction.

https://dx.doi.org/10.1021/acs.orglett.0c04191
Org. Lett. 2021, 23, 607−611

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c04191.

Experimental procedures, spectra for all new compounds, and X-ray crystallographic data for compounds **3e**, (±)-**5e**, and (±)-**5f** (PDF)

Accession Codes

CCDC 1437751−1437753 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Authors

Zhi-Tao He − *CAS Key Laboratory of Synthetic Chemistry of Natural Substances, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200032, China;* orcid.org/0000-0003-2375-9557; Email: hezt@sioc.ac.cn

Ping Tian − *The Research Center of Chiral Drugs, Innovation Research Institute of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China; CAS Key Laboratory of Synthetic Chemistry of Natural Substances, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200032, China;* orcid.org/0000-0002-5612-0664; Email: tianping@sioc.ac.cn, tianping@shutcm.edu.cn

Guo-Qiang Lin − *The Research Center of Chiral Drugs, Innovation Research Institute of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China; CAS Key Laboratory of Synthetic Chemistry of Natural Substances, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200032, China*; Email: lingq@sioc.ac.cn

Authors

Kai-Rui Feng − *The Research Center of Chiral Drugs, Innovation Research Institute of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China*

Yun-Xuan Tan − *The Research Center of Chiral Drugs, Innovation Research Institute of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China; CAS Key Laboratory of Synthetic Chemistry of Natural Substances, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200032, China;* orcid.org/0000-0002-5114-0359

Wenbo Ye − *The Research Center of Chiral Drugs, Innovation Research Institute of Traditional Chinese Medicine, Shanghai University of Traditional Chinese Medicine, Shanghai 201203, China*

Yi-Fan Wang − *CAS Key Laboratory of Synthetic Chemistry of Natural Substances, Shanghai Institute of Organic Chemistry, University of Chinese Academy of Sciences, Chinese Academy of Sciences, Shanghai 200032, China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.0c04191

Author Contributions

#K.-R.F. and Y.-X.T. contributed equally.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Financial support was generously provided by the National Natural Science Foundation of China (21871184, 22071155, and 21871284), the Shanghai Municipal Education Commission (2019-01-07-00-10-E00072), the Program of Shanghai Academic/Technology Research Leader (20XD1403600), the Science and Technology Commission of Shanghai Municipality (18401933500 and 20400750300), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDB 20020100), and the Key Research Program of Frontier Science (QYZDY-SSW-SLH026).

■ REFERENCES

(1) (a) Schulte, G.; Scheuer, P. J.; McConnell, O. J. Upial, a sesquiterpenoid bicyclo[3.3.1]nonane aldehyde lactone from the marine sponge Dysidea fragilis. *J. Org. Chem.* **1980**, *45*, 552. (b) Izumikawa, M.; Satou, R.; Motohashi, K.; Nagai, A.; Ohnishi, Y.; Takagi, M.; Shin-ya, K. Naphthoquinone-like Polyketide Isolated from Streptomyces sp. RI-77 and Its Predicted Biosynthetic Pathway. *J. Nat. Prod.* **2011**, *74*, 2588. (c) Krohn, K.; Biele, C.; Drogies, K.-H.; Steingröver, K.; Aust, H.-J.; Draeger, S.; Schulz, B. Fusidilactones, a New Group of Polycyclic Lactones from an Endophyte, Fusidium sp. *Eur. J. Org. Chem.* **2002**, *2002*, 2331. (d) Chen, Y.; Gan, F.; Jin, S.; Liu, H.; Wu, S.; Yang, W.; Yang, G. Adamantyl derivatives and rearranged benzophenones from Garcinia xanthochymus fruits. *RSC Adv.* **2017**, *7*, 17289. (e) Ishibashi, M.; Murae, T.; Hirota, H.; Tsuyuki, T.; Takahashi, T.; Itai, A.; Iitaka, Y. Shinjulactone C, a new quassinoid with a 1α, 12α:5α, 13α-dicyclo-9βH-picrasane skeleton from ailanthus altissima SWINGLE. *Tetrahedron Lett.* **1982**, *23*, 1205.

(2) For selected reviews on borylcopper-catalyzed borylation of allenes, see: (a) Pulis, A. P.; Yeung, K.; Procter, D. J. Enantioselective copper catalysed, direct functionalisation of allenes via allyl copper intermediates. *Chem. Sci.* **2017**, *8*, 5240. (b) Fujihara, T.; Tsuji, Y. Cu-Catalyzed Borylative and Silylative Transformations of Allenes: Use of β-Functionalized Allyl Copper Intermediates in Organic Synthesis. *Synthesis* **2018**, *50*, 1737. For selected recent reviews on borylcopper-catalyzed reactions, see: (c) Talbot, F. J. T.; Dherbassy, Q.; Manna, S.; Shi, C.; Zhang, S.; Howell, G. P.; Perry, G. J. P.; Procter, D. J. Copper-Catalyzed Borylative Couplings with C−N Electrophiles. *Angew. Chem., Int. Ed.* **2020**, *59*, 20278. (d) Whyte, A.; Torelli, A.; Mirabi, B.; Zhang, A.; Lautens, M. Copper-Catalyzed Borylative Difunctionalization of π-Systems. *ACS Catal.* **2020**, *10*, 11578. (e) Xue, W.; Oestreich, M. Beyond Carbon: Enantioselective and Enantiospecific Reactions with Catalytically Generated Boryl- and Silylcopper Intermediates. *ACS Cent. Sci.* **2020**, *6*, 1070.

(3) (a) Meng, F.; Jang, H.; Jung, B.; Hoveyda, A. H. Cu-Catalyzed Chemoselective Preparation of 2-(Pinacolato)boron-Substituted Allylcopper Complexes and their In Situ Site-, Diastereo-, and Enantioselective Additions to Aldehydes and Ketones. *Angew. Chem., Int. Ed.* **2013**, *52*, 5046. (b) Zhao, Y.-S.; Tang, X.-Q.; Tao, J.-C.; Tian, P.; Lin, G.-Q. Efficient access to *cis*-decalinol frameworks: copper(I)-catalyzed borylative cyclization of allene cyclohexanediones. *Org. Biomol. Chem.* **2016**, *14*, 4400. (c) Rae, J.; Yeung, K.; McDouall, J. J. W.; Procter, D. J. Copper-Catalyzed Borylative Cross-Coupling of Allenes and Imines: Selective Three-Component Assembly of Branched Homoallyl Amines. *Angew. Chem., Int. Ed.* **2016**, *55*, 1102. (d) Yeung, K.; Ruscoe, R. E.; Rae, J.; Pulis, A. P.; Procter, D. J. Enantioselective Generation of Adjacent Stereocenters in a Copper-

Catalyzed Three-Component Coupling of Imines, Allenes, and Diboranes. *Angew. Chem., Int. Ed.* **2016**, *55*, 11912. (e) Jang, H.; Romiti, F.; Torker, S.; Hoveyda, A. H. Catalytic diastereo- and enantioselective additions of versatile allyl groups to N−H ketimines. *Nat. Chem.* **2017**, *9*, 1269. (f) Deng, H.; Meng, Z.; Wang, S.; Zhang, Z.; Zhang, Y.; Shangguan, Y.; Yang, F.; Yuan, D.; Guo, H.; Zhang, C. Enantioselective Copper-Catalyzed Three-Component Carboboronation of Allenes: Access to Functionalized Dibenzo [*b*,*f*][1,4]oxazepine Derivatives. *Adv. Synth. Catal.* **2019**, *361*, 3582. (g) Yeung, K.; Talbot, F. J. T.; Howell, G. P.; Pulis, A. P.; Procter, D. J. Copper-Catalyzed Borylative Multicomponent Synthesis of Quaternary α-Amino Esters. *ACS Catal.* **2019**, *9*, 1655. (h) Zhao, C.-Y.; Zheng, H.; Ji, D.-W.; Min, X.-T.; Hu, Y.-C.; Chen, Q.-A. Copper-Catalyzed Asymmetric Carboboronation of Allenes to Access α-Quaternary Amino Esters with Adjacent Stereocenters. *Cell Reports Physical Science* **2020**, *1*, 100067. (i) Zhang, S.; del Pozo, J.; Romiti, F.; Mu, Y.; Torker, S.; Hoveyda, A. H. Delayed catalyst function enables direct enantioselective conversion of nitriles to NH_2-amines. *Science* **2019**, *364*, 45.

(4) (a) Meng, F.; McGrath, K. P.; Hoveyda, A. H. Multifunctional organoboron compounds for scalable natural product synthesis. *Nature* **2014**, *513*, 367. (b) Semba, K.; Bessho, N.; Fujihara, T.; Terao, J.; Tsuji, Y. Copper-Catalyzed Borylative Allyl-Allyl Coupling Reaction. *Angew. Chem., Int. Ed.* **2014**, *53*, 9007. (c) Xu, G.; Fu, B.; Zhao, H.; Li, Y.; Zhang, G.; Wang, Y.; Xiong, T.; Zhang, Q. Enantioselective and site-specific copper-catalyzed reductive allyl-allyl cross-coupling of allenes. *Chem. Sci.* **2019**, *10*, 1802.

(5) Zhou, Y.; You, W.; Smith, K. B.; Brown, M. K. Copper-Catalyzed Cross-Coupling of Boronic Esters with Aryl Iodides and Application to the Carboboration of Alkynes and Allenes. *Angew. Chem., Int. Ed.* **2014**, *53*, 3475.

(6) (a) Meng, F.; Li, X.; Torker, S.; Shi, Y.; Shen, X.; Hoveyda, A. H. Catalytic enantioselective 1,6-conjugate additions of propargyl and allyl groups. *Nature* **2016**, *537*, 387. (b) Huang, Y.; Torker, S.; Li, X.; del Pozo, J.; Hoveyda, A. H. Racemic Vinylallenesin Catalytic Enantioselective Multicomponent Processes: Rapid Generation of Complexity through 1,6-Conjugate Additions. *Angew. Chem., Int. Ed.* **2019**, *58*, 2685.

(7) (a) Fujihara, T.; Sawada, A.; Yamaguchi, T.; Tani, Y.; Terao, J.; Tsuji, Y. Boraformylation and Silaformylation of Allenes. *Angew. Chem., Int. Ed.* **2017**, *56*, 1539. (b) Boreux, A.; Indukuri, K.; Gagosz, F.; Riant, O. Acyl Fluorides as Efficient Electrophiles for the Copper-Catalyzed Boroacylation of Allenes. *ACS Catal.* **2017**, *7*, 8200. (c) Sawada, A.; Fujihara, T.; Tsuji, Y. Copper-Catalyzed Bora-Acylation and Bora-Alkoxyoxalylation of Allenes. *Adv. Synth. Catal.* **2018**, *360*, 2621. (d) Han, J.; Zhou, W.; Zhang, P.-C.; Wang, H.; Zhang, R.; Wu, H.-H.; Zhang, J. Design and Synthesis of WJ-Phos, and Application in Cu-Catalyzed Enantioselective Boroacylation of 1,1-Disubstituted Allenes. *ACS Catal.* **2019**, *9*, 6890.

(8) Gong, T.-J.; Yu, S.-H.; Li, K.; Su, W.; Lu, X.; Xiao, B.; Fu, Y. Copper-Catalyzed Alkynylboration of Alkenes with Diboron Reagents and Bromoalkynes. *Chem. - Asian J.* **2017**, *12*, 2884.

(9) (a) Kageyuki, I.; Osaka, I.; Takaki, K.; Yoshida, H. Copper-Catalyzed B(dan)-Installing Carboboration of Alkenes. *Org. Lett.* **2017**, *19*, 830. (b) Ozawa, Y.; Iwamoto, H.; Ito, H. Copper(I)-catalysed regio- and diastereoselective intramolecular alkylboration of terminal allenes via allylcopper(I) isomerization. *Chem. Commun.* **2018**, *54*, 4991.

(10) For 1,4-addition of β-borylated allylcopper intermediates, see: (a) Li, X.; Meng, F.; Torker, S.; Shi, Y.; Hoveyda, A. H. Catalytic Enantioselective Conjugate Additions of (pin)B-Substituted Allylcopper Compounds Generated in situ from Butadiene or Isoprene. *Angew. Chem., Int. Ed.* **2016**, *55*, 9997. For other recent copper-catalyzed borylative conjugate addition reactions, see: (b) Kim-Lee, S.-H.; Alonso, I.; Mauleón, P.; Arrayás, R. G.; Carretero, J. C. Rationalizing the Role of NaO*t*Bu in Copper-Catalyzed Carboboration of Alkynes: Assembly of Allylic All-Carbon Quaternary Stereocenters. *ACS Catal.* **2018**, *8*, 8993. (c) He, C.-Y.; Li, Q.-H.; Wang, X.; Wang, F.; Tian, P.; Lin, G.-Q. Copper-Catalyzed Asymmetric Borylative Cyclization of Cyclohexadienone-Containing 1,6-Dienes. *Adv. Synth. Catal.* **2020**, *362*, 765.

(11) For selected transition-metal-catalyzed cascade reactions of cyclohexadienone-tethered allene (1,6-enallene) substrates, see: (a) He, Z.-T.; Tang, X.-Q.; Xie, L.-B.; Cheng, M.; Tian, P.; Lin, G.-Q. Efficient Access to Bicyclo[4.3.0]nonanes: Copper-Catalyzed Asymmetric Silylative Cyclization of Cyclohexadienone-Tethered Allenes. *Angew. Chem., Int. Ed.* **2015**, *54*, 14815. (b) Tan, Y.-X.; Tang, X.-Q.; Liu, P.; Kong, D.-S.; Chen, Y.-L.; Tian, P.; Lin, G.-Q. CuH-Catalyzed Asymmetric Intramolecular Reductive Coupling of Allenes to Enones. *Org. Lett.* **2018**, *20*, 248. (c) Nguyen, T. L. N.; Incerti-Pradillos, C. A.; Lewis, W.; Lam, H. W. Enantioselective nickel-catalyzed arylative intramolecular 1,4-allylations. *Chem. Commun.* **2018**, *54*, 5622. (d) Kong, D.-S.; Wang, Y.-F.; Tan, Y.-X.; Zhang, J.-L.; Zhang, J.-G.; Tian, P.; Lin, G.-Q. Trisannulation of benzamides and cyclohexadienone-tethered 1,1-disubstituted allenes initiated by Cp*Rh(III)-catalyzed C−H activation. *Org. Chem. Front.* **2019**, *6*, 699.

(12) The conjugate borylation to 4,4-dimethyl-2-cyclohexenone was inhibited; see: Feng, X.; Yun, J. Catalytic enantioselective boron conjugate addition to cyclic carbonyl compounds: a new approach to cyclic β-hydroxy carbonyls. *Chem. Commun.* **2009**, *43*, 6577.

(13) Semba, K.; Fujihara, T.; Terao, J.; Tsuji, Y. Copper-Catalyzed Borylation of α-Alkoxy Allenes with Bis(pinacolato)diboron: Efficient Synthesis of 2-Boryl 1,3-Butadienes. *Angew. Chem., Int. Ed.* **2013**, *52*, 12400.

(14) For details on more chiral ligand screening, see the Supporting Information.

(15) Selected examples on conjugate borylation: (a) Lee, J.-E.; Yun, J. Catalytic Asymmetric Boration of Acyclic α,β-Unsaturated Esters and Nitriles. *Angew. Chem., Int. Ed.* **2008**, *47*, 145. (b) Chen, I.-H.; Yin, L.; Itano, W.; Kanai, M.; Shibasaki, M. Catalytic Asymmetric Synthesis of Chiral Tertiary Organoboronic Esters through Conjugate Boration of β-Substituted Cyclic Enones. *J. Am. Chem. Soc.* **2009**, *131*, 11664. (c) O'Brien, J. M.; Lee, K.-S.; Hoveyda, A. H. Enantioselective Synthesis of Boron-Substituted Quaternary Carbons by NHC−Cu-Catalyzed Boronate Conjugate Additions to Unsaturated Carboxylic Esters, Ketones, or Thioesters. *J. Am. Chem. Soc.* **2010**, *132*, 10630. (d) Moure, A. L.; Arrayás, R. G.; Carretero, J. C. Catalytic asymmetric conjugate boration of α,β-unsaturated sulfones. *Chem. Commun.* **2011**, *47*, 6701. (e) Kobayashi, S.; Xu, P.; Endo, T.; Ueno, M.; Kitanosono, T. Chiral Copper(II)-Catalyzed Enantioselective Boron Conjugate Additions to α,β-Unsaturated Carbonyl Compounds in Water. *Angew. Chem., Int. Ed.* **2012**, *51*, 12763. (f) He, Z.-T.; Zhao, Y.-S.; Tian, P.; Wang, C.-C.; Dong, H.-Q.; Lin, G.-Q. Copper-Catalyzed Asymmetric Hydroboration of α-Dehydroamino Acid Derivatives: Facile Synthesis of Chiral β-Hydroxy-α-amino Acids. *Org. Lett.* **2014**, *16*, 1426.

(16) Liu, P.; Fukui, Y.; Tian, P.; He, Z.-T.; Sun, C.-Y.; Wu, N.-Y.; Lin, G.-Q. Cu-Catalyzed Asymmetric Borylative Cyclization of Cyclohexadienone-Containing 1,6-Enynes. *J. Am. Chem. Soc.* **2013**, *135*, 11700.

(17) Li, Q.-H.; Gao, D.; He, C.-Y.; Liao, Q.; Tan, Y.-X.; Wang, Y.-H.; Ding, R.; Lin, G.-Q.; Tian, P. One-Pot Preparation of Bridged Tricyclic and Fused Tetracyclic Scaffolds via Rhodium(III)-Catalyzed Asymmetric Borylative Cyclization. *Cell Reports Physical Science* **2020**, *1*, 100222.

https://dx.doi.org/10.1021/acs.orglett.0c04191
Org. Lett. 2021, 23, 607−611